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Magnetic sensor half-bridge based on inverse spin hall effect with reduced thermal drift

Abstract

The present disclosure generally relates to a magnetic sensor half bridge circuit. The half bridge circuit comprises a bias source connected to a first leg and a second leg. The first leg comprises one or more first spin-orbit torque (SOT) structures connected in series, each first SOT structure comprising a first ferromagnetic (FM) layer disposed on a first SOT layer. The first SOT layer has a first end connected to ground and a second end connected to a first voltage sensor. The second leg comprises one or more second SOT structures connected in series, each second SOT structure comprising a second FM layer disposed on a second SOT layer. The second SOT layer has a first end connected to a second voltage sensor and a second end connected to a ground. The first voltage sensor is disposed adjacent to the second voltage sensor.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

(1) Embodiments of the present disclosure generally relate to a magnetic sensor half bridge.

Description of the Related Art

(2) A Wheatstone bridge is an electrical circuit used to measure an unknown electrical resistance by balancing two legs of a bridge circuit, one leg of which includes an unknown component. The Wheatstone circuit provides extremely accurate measurements in comparison to a simple voltage divider.

(3) The Wheatstone bridge includes multiple resistors that, especially recently, include magnetic material such as a magnetic field sensors where its resistance changes with the magnetic field.

Magnetic sensors can include Hall effect magnetic sensors, anisotropy magnetoresistive sensors (AMR), giant magnetoresistive (GMR) sensors, and tunnel magnetoresistive (TMR) sensors.

(4) The Wheatstone bridge has a linear output signal and resists the environment temperature fluctuations. Any temperature change in the Wheatstone bridge array is cancelled. The Wheatstone bridge array has four resistors with one resistor in each leg. Two of the resistors have identical resistance while the remaining two resistors have identical resistances relative to each other, but different from the original two resistors.

(5) It is common practice to build a Wheatstone bridge on a single chip with each leg of the bridge comprising one or more sensors to obtain a higher signal output while minimizing the thermal drift and achieving zero offset. In operation, the bridge uses opposite sensor field responses for adjacent legs (i.e., one leg resistance increases with field while sensors on the adjacent leg produces a reduction of resistance with an externally applied magnetic field. For conventional GMR or TMR based bridge sensors, opposite pinning direction is used between adjacent legs. This can generally lead to significant changes in designs/processes.

(6) The Wheatstone bridge has four legs as noted above. The four legs can take up a significant amount of space. As devices continue to shrink, the bridge needs to shrink as well.

(7) Therefore, there is a need in the art for an improved bridge.

SUMMARY OF THE DISCLOSURE

(8) The present disclosure generally relates to a magnetic sensor half bridge circuit. The half bridge circuit comprises a bias source connected to a first leg and a second leg. The first leg comprises one or more first spin-orbit torque (SOT) structures connected in series, each first SOT structure comprising a first ferromagnetic (FM) layer disposed on a first SOT layer. The first SOT layer has a first end connected to ground and a second end connected to a first voltage sensor. The second leg comprises one or more second SOT structures connected in series, each second SOT structure comprising a second FM layer disposed on a second SOT layer. The second SOT layer has a first end connected to a second voltage sensor and a second end connected to a ground. The first voltage sensor is disposed adjacent to the second voltage sensor.

(9) In one embodiment, a half bridge circuit comprises a first spin-orbit torque (SOT) layer having a first end and a second end opposite the first end, a first ferromagnetic (FM) layer disposed on the first SOT layer, a first ground connected to the first end of the first SOT layer, a first voltage sensor connected to the second end of the first SOT layer, a bias source connected to the first FM layer, a second SOT layer having a first end and a second end opposite the first end, a second FM layer disposed on the second SOT layer, the second FM layer being connected to the bias source, a second ground connected to the second end of the second SOT layer, and a second voltage sensor

connected to the first end of the second SOT layer.

(10) In another embodiment, a half bridge circuit comprises a bias source, a first leg connected to the bias source, the first leg comprising one or more first spin-orbit torque (SOT) structures, each first SOT structure comprising: a first SOT layer having a first end and a second end opposite the first end, a first ferromagnetic (FM) layer disposed on the first SOT layer, a first ground connected to the first end of the first SOT layer, and a first voltage sensor connected to the second end of the first SOT layer, and a second leg connected to the bias source, the second leg comprising one or more second SOT structures, each second SOT structure comprising: a second SOT layer having a first end and a second end opposite the first end, a second FM layer disposed on the second SOT layer, a second ground connected to the second end of the second SOT layer, and a second voltage sensor connected to the first end of the second SOT layer.

(11) In yet another embodiment, a half bridge circuit comprises a bias source, a first leg connected to the bias source, the first leg comprising one or more first spin-orbit torque (SOT) structures connected in series, each first SOT structure comprising: a first SOT layer having a first end and a second end opposite the first end, a first ferromagnetic (FM) layer disposed on the first SOT layer, a second FM layer disposed below the first SOT layer, a first ground connected to the first end of the first SOT layer, and a first voltage sensor connected to the second end of the first SOT layer, and a second leg connected to the bias source, the second leg comprising one or more second SOT structures connected in series, each second SOT structure comprising: a second SOT layer having a first end and a second end opposite the first end, wherein the first end of the second SOT layer is disposed adjacent to the second end of the first SOT layer, a third FM layer disposed on the second SOT layer, a fourth FM layer disposed below the second SOT layer, a second ground connected to the second end of the second SOT layer, and a second voltage sensor connected to the first end of the second SOT layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

(2) FIG. 1 is a schematic illustration of a Wheatstone bridge array design.

(3) FIGS. 2A and 2B are schematic illustrations of a half bridge circuit diagram, according to one embodiment.

(4) FIGS. 3A and 3B are schematic illustrations of legs of the half bridge design of FIG. 2B according to one embodiment.

(5) FIG. 4A is a schematic cross section illustration of a single ferromagnetic (FM) layer bridge according to one embodiment.

(6) FIG. 4B is a schematic top view of FIG. 4A.

(7) FIG. 4C is a schematic side view of a leg having an array of sensors according to one embodiment.

(8) FIG. 5 is a schematic cross-sectional view illustration of a half bridge design according to another embodiment.

(9) FIG. 6A is a schematic cross section illustration of a single ferromagnetic (FM) layer bridge according to one embodiment.

(10) FIG. 6B is a schematic cross-sectional view of a leg having an array of sensors according to

one embodiment.

(11) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

DETAILED DESCRIPTION

(12) In the following, reference is made to embodiments of the disclosure. However, it should be understood that the disclosure is not limited to specific described embodiments. Instead, any combination of the following features and elements, whether related to different embodiments or not, is contemplated to implement and practice the disclosure. Furthermore, although embodiments of the disclosure may achieve advantages over other possible solutions and/or over the prior art, whether or not a particular advantage is achieved by a given embodiment is not limiting of the disclosure. Thus, the following aspects, features, embodiments and advantages are merely illustrative and are not considered elements or limitations of the appended claims except where explicitly recited in a claim(s). Likewise, reference to “the disclosure” shall not be construed as a generalization of any inventive subject matter disclosed herein and shall not be considered to be an element or limitation of the appended claims except where explicitly recited in a claim(s).

(13) The present disclosure generally relates to a magnetic sensor half bridge circuit. The half bridge circuit comprises a bias source connected to a first leg and a second leg. The first leg comprises one or more first spin-orbit torque (SOT) structures connected in series, each first SOT structure comprising a first ferromagnetic (FM) layer disposed on a first SOT layer. The first SOT layer has a first end connected to ground and a second end connected to a first voltage sensor. The second leg comprises one or more second SOT structures connected in series, each second SOT structure comprising a second FM layer disposed on a second SOT layer. The second SOT layer has a first end connected to a second voltage sensor and a second end connected to a ground. The first voltage sensor is disposed adjacent to the second voltage sensor.

(14) FIG. 1 is a schematic illustration of a Wheatstone bridge array **100** design. The array **100** includes a bias source **102**, a first resistor **104**, a second resistor **106**, a third resistor **108**, a fourth resistor **110**, a first sensing port **112**, a second sensing port **114**, and a ground connection **116**. Bias voltage is applied across the array from the bias source **102** to the ground connection **116**. The first sensing port **112** and the second sensing port **114** sense the output of the applied voltage. Any temperature change from the resistors **104**, **106**, **108**, **110** can be cancelled.

(15) As discussed herein, the resistors **104**, **106**, **108**, **110** each include a sensor, such as a tunneling magnetoresistance (TMR) sensor. The TMR sensors may each be distinct and different such that the resistors **104**, **106**, **108**, **110** have different resistance. The sensors may be identical, but the resistors **104**, **106**, **108**, **110** are different. Resistors **104**, **110** may be identical to each other (as are the TMR sensors that comprise the resistors **104**, **110**), and resistors **106**, **108** are identical to each other (as are the sensors that comprise the resistors **106**, **108**) yet different from resistors **104**, **110**. For a TMR sensor in array **100**, the RA for the array **100** is around 100 Ohms microns.^{sup.2}

(16) Typical magnetic field sensors use MR (magnetoresistance) devices, like TMR devices, in a Wheatstone bridge circuit. The sensor requires the MR devices to change differently in adjacent legs of the bridge. One way to make a magnetic field sensor is to fabricate two different TMR films with opposite pinning directions on the in the same wafer and anneal wafers in multiple temperatures/field conditions. The reliability and performance of the TMR films determines the magnetoresistance response. In this way, combined with different TMR films features, a perfect Wheatstone bridge design for magnetic field sensor can be fabricated. However, it will introduce significant amounts of wafer/device processing complexity.

(17) FIGS. 2A-2B are a schematic illustrations of a half bridge circuit diagram **200** of a magnetic sensor, according to one embodiment. The half bridge circuit **200** may be referred to herein as a magnetic sensor **200**.

(18) The half bridge circuit **200** comprises a first leg **205** comprising a first three-terminal spin-orbit torque (SOT) resistor (**R1**) **204** connected to a bias source (**Vdd**) **202**, a first voltage sensor (**Vdc1**) **220**, and to ground **216**. The half bridge circuit **200** further comprises a second leg **207** comprising a second three-terminal SOT resistor (**R2**) **206** connected to the bias source **202**, a second voltage sensor (**Vdc2**) **222**, and to ground **216**. The first and second resistors **204**, **206** may be connected to different ground sources **216a**, **216b**, like shown in FIG. 2B. In such embodiments, the first leg **205** comprises the first resistor **204**, the ground **216a**, and the first voltage sensor **220**, and the second leg **207** comprises the second resistor **206**, the ground **216b**, and the second voltage sensor **222**. The first and second resistors **204**, **206** are connected to one another in series. The resistors **204**, **206** may be referred to as SOT structures. All elements and components of the half bridge circuit **200** can be deposited and fabricated on the same wafer, avoiding the complexity as described in TMR sensor based full Wheatstone bridges.

(19) As shown in FIG. 2B, the first resistor **204** comprises an SOT layer **224a** and a ferromagnetic (FM) layer **226**. A first end **223a** of the SOT layer **224a** is connected to ground **216a**, and a second end **223b** of the SOT layer **224a** is connected to the first voltage sensor **220**. The FM layer **226** is disposed on the SOT layer **224** (with possible intervening layers not shown, as will be the case for the following figures) and connected to the bias source **202**. The second resistor **206** comprises an SOT layer **228** and an FM layer **230**. A first end **225a** of the SOT layer **228** is connected to the second voltage sensor **222**, and a second end **225b** of the SOT layer **228** is connected to ground **216b**. The FM layer **230** is disposed on the SOT layer **228** (with possible intervening layers not shown, as will be the case for the following figures) and connected to the bias source **202**. The FM layers **226**, **230** are stabilized such that the FM layers **226**, **230** are magnetized in the x-direction. More generally, a layer described as being disposed on another layer does not require the two layers to be in contact, and one or more intervening layers may be present in between.

(20) The second end **223a** of the SOT layer **224** of the first resistor **204** is disposed adjacent to the first end **225a** of the SOT layer **228** of the second resistor **206** such that the first voltage sensor **220** is disposed adjacent to the second voltage sensor **222**. As such, the ground **216a** and the ground **216b** are disposed on opposite ends of the SOT layers **224**, **228**. The SOT layers **224** and **228** are substantially the same, and the FM layers **226** and **230** are substantially the same. This enables the same deposition of whole stack (FM/SOT) all at once and fabrication on the same wafer.

(21) Each FM layer **226**, **230** has a width **232** in the x-direction and a height **233** in the y-direction. The width **232** may be greater than the height **233** to utilize shape anisotropy of the FM layers **226**, **230**. The aspect ratio (width divided by height) is generally greater than or equal to 3. For example, the width **232** may be about 300 nm to about 30 μ m and the height **233** may be about 100 nm to about 10 μ m. While the FM layers **226**, **230** are shown as rectangular, the FM layers **226**, **230** may instead be elliptical in shape, like discussed below in FIG. 4B.

(22) The FM layers **226**, **230** may each individually comprise NiFe, CoFe, NiFeX, CoFeX, FeX, or Nix, where X=Co, Ni, Cu, Si, Al, Mn, Ge, Ta, Hf, N, and B. The FM layers **226**, **230** may comprise any magnetic layer combination or alloy combination of these elements that can yield a low coercivity, negative magnetostrictive FM layers **226**, **230** or in multilayer combinations with other higher polarizing materials like Heusler alloys or high Ni containing alloy FM layers.

(23) The SOT layers **224**, **228** may each individually comprise doped or undoped bismuth antimony (BiSb) of various thicknesses. In one embodiment, the SOT layers **224**, **228** comprises undoped BiSb, or doped BiSbX, where X is less than at. 8% and extracted from elements which don't readily interact with either Bi or Sb, such as Cu, Ag, Ge, Mg, Ni, Co Mo, W, Sn, B, N, In, Te, Se, Y, Zr, Pt, Ti, or in alloy combinations thereof, or as ceramic oxide, carbide, or nitride dopants of these elements. The SOT layers **224**, **228** may each individually comprise YPtBi.

(24) The bias source **202** provides a perpendicular current passing through the FM layers **226**, **230**, getting spin polarized and injecting into the SOT layers **224**, **228** of the first and second resistors **202**, **204** (along the z-direction). Due to the inverse spin Hall effect (iSHE), a DC voltage is

induced between the ground **216a**, **216b** and the first and second voltage sensors **220**, **222** longitudinally (along the x-direction). The magnitude of the DC voltage depends on the magnetization components of the FM layers **226**, **230** in the y-direction. If the magnetization of the FM layers **226**, **230** is 100% along the y-direction, the two DC voltages (V_{dc1} and V_{dc2}) will have a maximum signal. If the magnetization of the FM layers **226**, **230** is 100% along the x-direction, the two DC voltages will have zero induced signal.

(25) When an external magnetic field is applied in the y-direction, the magnetizations of the FM layers **226**, **230** will rotate such that at the first resistor **204**, the first voltage sensor **220** is greater than 0, and at the second resistor **206**, the second voltage sensor **222** is less than 0 (or vice versa), resulting in the signal ($V_{dc2}-V_{dc1}$) being doubled. When no external magnetic field is applied or an external field is applied along the x-direction, the magnetizations of the FM layers **226**, **230** remain in the x-direction, resulting in no intrinsic signal output besides some thermal drift, as the first and second voltage sensors **220**, **222** are equal ($V_{dc1}=V_{dc2}$). This way, the half bridge design **200** is only sensitive to the magnetic field along the y-direction and immune to magnetic fields in any other components.

(26) The half bridge design **200** comprising only two legs **205**, **207** is able to achieve the same functionality of the Wheatstone bridge array **100** of FIG. 1 in a more simplistic and straightforward manner. Because the half bridge circuit **200** utilizes SOT structures without using the TMR effect, it is less likely to breakdown over time (intrinsic reliability concern for TMR sensors). Moreover, the output of the two legs **205**, **207** of the half bridge design **200** have opposite polarities, resulting in the signal of the half bridge design **200** to be doubled or boosted. Additionally, any environment/thermal drifts will cause the same changes (both magnitude and polarity) of the V_{dc1} and V_{dc2} , resulting in net cancelation of the signal output.

(27) FIGS. 3A and 3B are schematic illustrations of the legs **205**, **207** of the half bridge design **200** of FIG. 2B, according to one embodiment. FIG. 3A illustrates the first leg **205**, and FIG. 3B illustrates the second leg **207**.

(28) As shown in FIG. 3A, the first leg **205** may comprise a plurality of SOT layers **224a**, **224b**, **224n** (collectively referred to herein as SOT layers **224**) with a plurality of FM layers **226a**, **226b**, **226n** (collectively referred to herein as FM layers **226**) disposed thereon. The plurality of SOT layers **224a**, **224b**, **224n** are connected together in series. Each FM layer **226a**, **226b**, **226n** is connected to the bias source **202**. One SOT layer **224** and the FM layer **226** disposed thereon may be referred to herein as an SOT structure. While three SOT structures are shown, the first leg **205** may comprise any number of SOT structures.

(29) The first SOT layer **224a** in the series has a first end **223a** connected to the ground **216a**. The last SOT layer **224n** in the series has a second end **223b** connected to the first voltage sensor **220**. A bottom surface **223c** of the last SOT layer **224n** opposite the surface the FM layer **226n** is disposed on is further coupled to a resistor **234b** and then to ground **236b**. Similarly, a bottom surface **223c** of each middle SOT layer(s) **224b** opposite the surface the FM layer **226b** is disposed on is coupled to a resistor **234a** and then to ground **236a**. Because the first SOT layer **224a** is connected to ground **216a** on the first end **223a**, the first SOT layer **224a** does not require a resistor like the middle and last SOT layers **224b**, **224n**. During operation, the voltage of each of the plurality of SOT layers **224a**, **224b**, **224n** and FM layers **226a**, **226b**, **226c** is added together, increasing the signal output. Moreover, all SOT layers **224a**, **224b**, **224n** and FM layers **226a**, **226b**, **226c** can be deposited and fabricated on the same thin wafer, as SOT layers **224a**, **224b**, **224n** are all the same, and the FM layers **226a**, **226b**, **226c** are all the same.

(30) Similar to the first leg **205**, the second leg **207** may comprise a plurality of SOT layers **228a**, **228b**, **228n** (collectively referred to herein as SOT layers **228**) with a plurality of FM layers **230a**, **230b**, **230n** (collectively referred to herein as SOT layers **230**) disposed thereon. The plurality of SOT layers **228a**, **228b**, **228n** are connected together in series. Each FM layer **230a**, **230b**, **230n** is connected to the bias source **202**. While three SOT layers **228a**, **228b**, **228n** and three FM layers

226a, 226b, 226b are shown, the second leg **207** may comprise any number of SOT layers and FM layers. One SOT layer **228** and the FM layer **230** disposed thereon may be referred to herein as an SOT structure.

(31) The last SOT layer **228n** in the series has a second end **225a** connected to the ground **216b**. The first SOT layer **228a** in the series has a first end **225a** connected to the second voltage sensor **222**. A bottom surface **225c** of the first SOT layer **228a** opposite the surface the FM layer **230a** is disposed on is further coupled to a resistor **234c** and then to ground **236d**. Similarly, a bottom surface **225c** of each middle SOT layer(s) **228b** opposite the surface the FM layer **228b** is disposed on is coupled to a resistor **234d** and then to ground **236d**. Because the last SOT layer **228n** is connected to ground **216b** on the second end **225b**, the last SOT layer **228n** does not require a resistor like the first and middle SOT layers **228a, 228b**. During operation, the voltage of each of the plurality of SOT layers **228a, 228b, 228n** and FM layers **230a, 230b, 230b** is added together, increasing the signal output.

(32) Moreover, all SOT layers **228a, 228b, 228n** and FM layers **230a, 230b, 230b** can be deposited and fabricated on the same wafer. Further, all SOT layers and FM layers of both the first and second legs **205, 207** can be deposited and fabricated on the same wafer. All SOT layers and FM layers of both the first and second legs **205, 207** may comprise any of the materials discussed above.

(33) FIG. 4A is a schematic cross section illustration of a single ferromagnetic (FM) layer bridge **405**, according to one embodiment. FIG. 4B is a schematic top view of FIG. 4A. The single FM layer bridge **405** may be either the first leg **205** or the second leg **207** of FIGS. 2A-3B. For clarity, ground, bias source, and voltage sensors are not shown in the FM layer bridge **405**. However, the FM layer bridge **405** may be connected as discussed above in FIGS. 2A-3B.

(34) The FM layer bridge **405** comprises an SOT layer **224** and an FM layer **226** disposed thereon (i.e., one SOT structure). The FM layer **226** is sandwiched between a first bias layer **440a** and a second bias layer **440b**. The first and second biasing layers **440a, 440b** may be either hard bias layers or soft bias layers. The first and second biasing layers **440a, 440b** may each individually comprise layers of Co, Pt, Ni, CoPt, CoPtCr, NiFe, CoFe, or combinations thereof. For the case of soft bias material, there is an additional AFM layer **442** disposed on top of the first and second biasing layers **440a, 440b** for exchanging bias of the soft bias layer. For case of hard bias layer, the AFM layer **442** can be skipped. The first and second biasing layers **440a, 440b** enable the FM layer **226** to have a magnetization in the x-direction when there is no external field applied. The strength of the first and second biasing layers **440a, 440b** controls the dynamic range of the overall sensor comprising the FM layer bridge **405**. An insulating layer may be disposed between the FM layer **226** and the biasing layers **440a, 440b** for electrical isolation. As shown in the side view of FIG. 4B (i.e., viewed from the z-direction), the FM layer **226** may be elliptical or oval in shape, and may utilize shape anisotropy when the width **232** is greater than the height **233**.

(35) FIG. 4C is a schematic cross-sectional view of a leg **407** having more than one SOT structure, according to one embodiment. For clarity, ground, bias source, and voltage sensors are not shown in the leg **407**. However, the leg **407** may be connected as discussed above in FIGS. 2A-3B. The leg **407** may be either the first or second leg **205, 207** of FIGS. 2A-3B.

(36) The leg **407** is similar to the legs **205, 207** shown in FIGS. 3A-3B. The leg **407** further comprises a first FM layer **226a** disposed over a first SOT layer **224a** and a second FM layer **226b** disposed over a second SOT layer **224b**. The first FM layer **226a** is disposed between a first biasing layer **440a** and a second biasing layer **440b**. The second FM layer **226b** is disposed between the second biasing layer **440b** and a third biasing layer **440c**. Thus, the first and second FM layers **226a, 226b** can share the second biasing layer **440b**, enabling the second biasing layer **440b** to bias both FM layers **226a, 226b**. While two FM layers **226a, 226b** and two SOT layers **224a, 224b** are shown, the leg **407** may comprise any number of FM layers, SOT layers, and biasing layers.

(37) FIG. 5 is a schematic illustration of a half bridge circuit design **500** of a magnetic sensor,

according to another embodiment. The half bridge circuit **500** may be referred to herein as a magnetic sensor **500**.

(38) The half bridge circuit **500** is similar to the half bridge circuit **200** of FIGS. 2A-2B such that the half bridge **500** design comprises a first leg **505** and a second leg **507**. The first leg **505** is similar to the first leg **205**, comprising the SOT layer **224** and a first FM layer (FM1) **226** disposed thereon. However, the first leg **505** further comprises a second FM layer (FM2) **526** disposed below the SOT layer **224** such that the SOT layer **224** is sandwiched between the first and second FM layers **226**, **526**. The first FM layer **226** is connected to the bias source **202**. The first end **223a** of the SOT layer **224** is connected to ground **216a**, and the second end **223b** of the SOT layer **224** is connected to the first voltage sensor **220**.

(39) Likewise, the second leg **507** is similar to the first leg **207**, comprising the SOT layer **228** and a first FM layer (FM1) **230** disposed thereon. However, the first leg **507** further comprises a second FM layer (FM2) **530** disposed below the SOT layer **228** such that the SOT layer **228** is sandwiched between the first and second FM layers **230**, **530**. The first FM layer **230** is connected to the bias source **202**. The first end **225a** of the SOT layer **228** is connected to ground **216b**, and the second end **225b** of the SOT layer **228** is connected to the second voltage sensor **222**.

(40) The second FM layers **526**, **530** may comprise the same materials as the first FM layers **226**, **230**, and have the same or different width **232** and height **233**. The first and second legs **505**, **507** may each individually comprise one or more SOT structures, like discussed above in FIGS. 3A-3B and shown in FIG. 6B. Moreover, the FM layers **226**, **526**, **230**, **530** may be elliptical in shape. All SOT layers **226**, **228** and FM layers **226**, **526**, **230**, **530** can be deposited and fabricated on the same thin wafer. Further, all SOT layers and FM layers of both the first and second legs **505**, **507** can be deposited and fabricated on the same thin wafer.

(41) The bias source **202** provides perpendicular current passing through the FM layers **226**, **230**, **526**, **530**, getting spin polarized current and injecting into the SOT layers **224**, **228** of the first and second legs **505**, **507** (along the z-direction). Due to the inverse spin Hall effect (iSHE), a DC voltage is induced between the ground **216a**, **216b** and the first and second voltage sensors **220**, **222** longitudinally (along the x-direction). The magnitude of the DC voltage depends on the magnetization of the FM layers **226**, **230**, **526**, **530** in the y-direction. In the half bridge design **500**, a first signal is induced at the interface between the first FM layer **226**, **230** and the SOT layers **224**, **228**, and a second signal is induced at the interface between the second FM layers **526**, **530** and the SOT layers **224**, **229**. The second voltage signal of the second FM layers **526**, **530** has the same polarity at the first voltage signal of the first FM layers **226**, **230**, and hence the first and second voltage signals can be added together because both the spin current direction and spin-polarization are reversed.

(42) When an external magnetic field is applied in the y-direction, the magnetizations of the FM layers **226**, **230**, **526**, **530** will rotate such that at the first leg **505**, the first voltage sensor **220** is greater than 0, and at the second leg **507**, the second voltage sensor **222** is less than 0 (or vice versa), resulting in the voltage difference being doubled yet again. When no external magnetic field is applied or the applied field is along the x-direction, the magnetizations of the FM layers **226**, **230**, **526**, **530** remain in the x-direction, resulting in no intrinsic signal output, as the first and second voltage sensors **220**, **222** are equal ($V_{dc1}=V_{dc2}$). This way, the half bridge design **500** is only sensitive to a magnetic field along the y-direction and immune to magnetic fields in any other components. Additionally, any environmental/thermal drifts will cause the same changes (both magnitude and polarity) of V_{dc1} and V_{dc2} , resulting in a net cancelation of the signal output.

(43) FIG. 6A is a schematic cross section illustration of a dual ferromagnetic (FM) layer bridge **605**, according to one embodiment. The dual FM layer bridge **605** may be either the first leg **505** or the second leg **507** of FIG. 5. For clarity, ground, bias source, and voltage sensors are not shown in the FM layer bridge **605**. However, the FM layer bridge **605** may be connected as discussed above in FIGS. 2A-3B and 5.

(44) The FM layer bridge **605** comprises a first FM layer **226**, a second FM layer **526**, and the SOT layer **224** disposed therebetween (i.e., one SOT structure). The first FM layer **226** is sandwiched between a first bias layer **440a** and a second bias layer **440b**. The second FM layer **526** is sandwiched between a third bias layer **640a** and a fourth bias layer **640b**. The first, second, third, and fourth biasing layers **440a**, **440b**, **640a**, **640b** may be either hard bias layers or soft bias layers. For the case of soft bias material, there are additional AFM layers **442**, **642** disposed on top of the first and second biasing layers **440a**, **440b**, and below the third and fourth biasing layers **640a**, **640b** for exchanging bias of the soft bias layer. For case of hard bias layers, the AFM layers **442**, **642** can be skipped. The first, second, third, and fourth biasing layers **440a**, **440b**, **640a**, **640b** may each individually comprise any material disclosed above for the first and second biasing layers **440a**, **440b**. The first and second biasing layers **440a**, **440b** enable the first FM layer **226** to have a magnetization in the x-direction when there is no external field applied, and the third and fourth biasing layers **640a**, **640b** enable the second FM layer **526** to have a magnetization in the x-direction when there is no external field applied. The strength of the first, second, third, and fourth biasing layers **440a**, **440b**, **640a**, **640b** controls the dynamic range of the overall sensor comprising the FM layer bridge **605**. An insulating layer may be disposed between the first FM layer **226** and the biasing layers **440a**, **440b**, and between the second FM layer **526** and the biasing layers **640a**, **640b**.

(45) FIG. **6B** is a schematic cross-sectional view or bottom view of a leg **607** having more than one SOT structure, according to one embodiment. For clarity, ground, bias source, and voltage sensors are not shown in the leg **607**. However, the leg **607** may be connected as discussed above in FIGS. **2A-3B** and **5**. The leg **607** may be either the first or second leg **505**, **507** of FIG. **5**.

(46) The leg **607** is similar to the legs **505**, **507** shown in FIG. **5**. The leg **607** comprises a first FM layer **226a**, a second FM layer **526a**, and a first SOT layer **224a** disposed therebetween, and a third FM layer **226b**, a fourth FM layer **526b**, and a second SOT layer **224b** disposed therebetween. The first FM layer **226a** is disposed between the first biasing layer **440a** and the second biasing layer **440b**. The third FM layer **226b** is disposed between the second biasing layer **440b** and a fifth biasing layer **440c**. Thus, the first and third FM layers **226a**, **226b** can share the second biasing layer **440b**, enabling the second biasing layer **440b** to bias both the first and third FM layers **226a**, **226b**.

(47) Similarly, the second FM layer **526a** is disposed between the third biasing layer **640a** and the fourth biasing layer **640b**. The fourth FM layer **526b** is disposed between the third biasing layer **640a** and a sixth biasing layer **640c**. Thus, the second and fourth FM layers **526a**, **526b** can share the fourth biasing layer **640b**, enabling the fourth biasing layer **640b** to bias both the second and fourth FM layers **526a**, **526b**. While two SOT structures are shown, the leg **607** may comprise any number of SOT structures.

(48) Therefore, a magnetic sensor comprising only two legs is able to achieve the same functionality of the Wheatstone bridge array in a more simplistic and straight-forward manner while further being less complex to fabricate. Furthermore, because the half bridge circuit utilizes SOT structures without using TMR effect, it is less likely to breakdown over time (intrinsic reliability concern for TMR sensors). Moreover, in the half bridge design, the output of the two legs have opposite polarities, resulting in the signal of the half bridge design being doubled or boosted. Additionally, any environmental/thermal drifts will cause the same changes (both magnitude and polarity) of V_{dc1} and V_{dc2} , resulting in a net cancelation of the drifts in the signal output.

(49) In one embodiment, a half bridge circuit comprises a first spin-orbit torque (SOT) layer having a first end and a second end opposite the first end, a first ferromagnetic (FM) layer disposed on the first SOT layer, a first ground connected to the first end of the first SOT layer, a first voltage sensor connected to the second end of the first SOT layer, a bias source connected to the first FM layer, a second SOT layer having a first end and a second end opposite the first end, a second FM layer

disposed on the second SOT layer, the second FM layer being connected to the bias source, a second ground connected to the second end of the second SOT layer, and a second voltage sensor connected to the first end of the second SOT layer.

(50) The first voltage sensor is disposed adjacent to the second voltage sensor. A ratio of a width of the first FM layer to a height of the first FM layer is greater than or equal to 3, and wherein a ratio of a width of the second FM layer to a height of the second FM layer is greater than or equal to 3. The half bridge circuit further comprises a first biasing layer disposed adjacent to the first FM layer over the first end of the first SOT layer, a second biasing layer adjacent to the first FM layer over the second end of the first SOT layer, a third biasing layer disposed adjacent to the second FM layer over the first end of the second SOT layer, and a fourth biasing layer disposed adjacent to the second FM layer over the second end of the second SOT layer. The half bridge circuit further comprises a third FM layer disposed below the first SOT layer, and a fourth FM layer disposed below the second SOT layer. The first FM layer and the second FM layer are each individually rectangular or elliptical in shape. A magnetic sensor comprising the half bridge circuit.

(51) In another embodiment, a half bridge circuit comprises a bias source, a first leg connected to the bias source, the first leg comprising one or more first spin-orbit torque (SOT) structures, each first SOT structure comprising: a first SOT layer having a first end and a second end opposite the first end, a first ferromagnetic (FM) layer disposed on the first SOT layer, a first ground connected to the first end of the first SOT layer, and a first voltage sensor connected to the second end of the first SOT layer, and a second leg connected to the bias source, the second leg comprising one or more second SOT structures, each second SOT structure comprising: a second SOT layer having a first end and a second end opposite the first end, a second FM layer disposed on the second SOT layer, a second ground connected to the second end of the second SOT layer, and a second voltage sensor connected to the first end of the second SOT layer.

(52) The first leg comprises a plurality of first SOT structures connected in series, and the second leg comprises a plurality of second SOT structures connected in series. The bias source is connected to the first FM layer and to the second FM layer. The first SOT layer and the second SOT layer each individually comprises BiSb or YPtSb. The second end of the first SOT layer is disposed adjacent to the first end of the second SOT layer. The first FM layer is disposed between a first biasing layer and a second biasing layer, and wherein the second FM layer is disposed between a third biasing layer and a fourth biasing layer. A magnetic sensor comprising the half bridge circuit.

(53) In yet another embodiment, a half bridge circuit comprises a bias source, a first leg connected to the bias source, the first leg comprising one or more first spin-orbit torque (SOT) structures connected in series, each first SOT structure comprising: a first SOT layer having a first end and a second end opposite the first end, a first ferromagnetic (FM) layer disposed on the first SOT layer, a second FM layer disposed below the first SOT layer, a first ground connected to the first end of the first SOT layer, and a first voltage sensor connected to the second end of the first SOT layer, and a second leg connected to the bias source, the second leg comprising one or more second SOT structures connected in series, each second SOT structure comprising: a second SOT layer having a first end and a second end opposite the first end, wherein the first end of the second SOT layer is disposed adjacent to the second end of the first SOT layer, a third FM layer disposed on the second SOT layer, a fourth FM layer disposed below the second SOT layer, a second ground connected to the second end of the second SOT layer, and a second voltage sensor connected to the first end of the second SOT layer.

(54) A ratio of a width of the first FM layer to a height of the first FM layer is greater than or equal to 3, a ratio of a width of the second FM layer to a height of the second FM layer is greater than or equal to 3, a ratio of a width of the third FM layer to a height of the third FM layer is greater than or equal to 3, and a ratio of a width of the fourth FM layer to a height of the fourth FM layer is greater than or equal to 3. The first FM layer is disposed between a first biasing layer and a second

biasing layer, the second FM layer is disposed between a third biasing layer and a fourth biasing layer, the third FM layer is disposed between a fifth biasing layer and a sixth biasing layer, and the fourth FM layer is disposed between a seventh biasing layer and an eighth biasing layer. The first, second, third, fourth, fifth, and sixth biasing layers are hard biasing layer. The first, second, third, fourth, fifth, and sixth biasing layers are soft biasing layer. A width of the first FM layer is greater than a height of the first FM layer, a width of the second FM layer is greater than a height of the second FM layer, a width of the third FM layer is greater than a height of the third FM layer, and a width of the fourth FM layer is greater than a height of the fourth FM layer. A magnetic sensor comprising the half bridge circuit.

(55) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A half bridge circuit, comprising: a first spin-orbit torque (SOT) layer having a first end and a second end opposite the first end; a first ferromagnetic (FM) layer disposed on the first SOT layer; a first ground connected to the first end of the first SOT layer; a first voltage sensor connected to the second end of the first SOT layer; a bias source connected to the first FM layer; a second SOT layer having a first end and a second end opposite the first end; a second FM layer disposed on the second SOT layer, the second FM layer being connected to the bias source; a second ground connected to the second end of the second SOT layer; and a second voltage sensor connected to the first end of the second SOT layer.
2. The half bridge circuit of claim 1, wherein the first voltage sensor is disposed adjacent to the second voltage sensor.
3. The half bridge circuit of claim 1, wherein a ratio of a width of the first FM layer to a height of the first FM layer is greater than or equal to 3, and wherein a ratio of a width of the second FM layer to a height of the second FM layer is greater than or equal to 3.
4. The half bridge circuit of claim 1, further comprising: a first biasing layer disposed adjacent to the first FM layer over the first end of the first SOT layer; a second biasing layer disposed adjacent to the first FM layer over the second end of the first SOT layer; a third biasing layer disposed adjacent to the second FM layer over the first end of the second SOT layer; and a fourth biasing layer disposed adjacent to the second FM layer over the second end of the second SOT layer.
5. The half bridge circuit of claim 1, further comprising: a third FM layer disposed below the first SOT layer; and a fourth FM layer disposed below the second SOT layer.
6. The half bridge circuit of claim 1, wherein the first FM layer and the second FM layer are each individually rectangular or elliptical in shape.
7. A magnetic sensor comprising the half bridge circuit of claim 1.
8. A half bridge circuit, comprising: a bias source; a first leg connected to the bias source, the first leg comprising one or more first spin-orbit torque (SOT) structures, each first SOT structure comprising: a first SOT layer having a first end and a second end opposite the first end; a first ferromagnetic (FM) layer disposed on the first SOT layer; a first ground connected to the first end of the first SOT layer; and a first voltage sensor connected to the second end of the first SOT layer; and a second leg connected to the bias source, the second leg comprising one or more second SOT structures, each second SOT structure comprising: a second SOT layer having a first end and a second end opposite the first end; a second FM layer disposed on the second SOT layer; a second ground connected to the second end of the second SOT layer; and a second voltage sensor connected to the first end of the second SOT layer.
9. The half bridge circuit of claim 8, wherein the first leg comprises a plurality of first SOT structures connected in series, and the second leg comprises a plurality of second SOT structures

connected in series.

10. The half bridge circuit of claim 8, wherein the bias source is connected to the first FM layer and to the second FM layer.

11. The half bridge circuit of claim 8, wherein the first SOT layer and the second SOT layer each individually comprises BiSb or YPtSb.

12. The half bridge circuit of claim 8, wherein the second end of the first SOT layer is disposed adjacent to the first end of the second SOT layer.

13. The half bridge circuit of claim 8, wherein the first FM layer is disposed between a first biasing layer and a second biasing layer, and wherein the second FM layer is disposed between a third biasing layer and a fourth biasing layer.

14. A magnetic sensor comprising the half bridge circuit of claim 8.

15. A half bridge circuit, comprising: a bias source; a first leg connected to the bias source, the first leg comprising one or more first spin-orbit torque (SOT) structures connected in series, each first SOT structure comprising: a first SOT layer having a first end and a second end opposite the first end; a first ferromagnetic (FM) layer disposed on the first SOT layer; a second FM layer disposed below the first SOT layer; a first ground connected to the first end of the first SOT layer; and a first voltage sensor connected to the second end of the first SOT layer; and a second leg connected to the bias source, the second leg comprising one or more second SOT structures connected in series, each second SOT structure comprising: a second SOT layer having a first end and a second end opposite the first end, wherein the first end of the second SOT layer is disposed adjacent to the second end of the first SOT layer; a third FM layer disposed on the second SOT layer; a fourth FM layer disposed below the second SOT layer; a second ground connected to the second end of the second SOT layer; and a second voltage sensor connected to the first end of the second SOT layer.

16. The half bridge circuit of claim 15, wherein: a ratio of a width of the first FM layer to a height of the first FM layer is greater than or equal to 3; a ratio of a width of the second FM layer to a height of the second FM layer is greater than or equal to 3; a ratio of a width of the third FM layer to a height of the third FM layer is greater than or equal to 3; and a ratio of a width of the fourth FM layer to a height of the fourth FM layer is greater than or equal to 3.

17. The half bridge circuit of claim 15, wherein: the first FM layer is disposed between a first biasing layer and a second biasing layer; the second FM layer is disposed between a third biasing layer and a fourth biasing layer; the third FM layer is disposed between a fifth biasing layer and a sixth biasing layer; and the fourth FM layer is disposed between a seventh biasing layer and an eighth biasing layer.

18. The half bridge circuit of claim 17, wherein the first, second, third, fourth, fifth, and sixth biasing layers are hard biasing layer.

19. The half bridge circuit of claim 17, wherein the first, second, third, fourth, fifth, and sixth biasing layers are soft biasing layer.

20. The half bridge circuit of claim 15, wherein: a width of the first FM layer is greater than a height of the first FM layer; a width of the second FM layer is greater than a height of the second FM layer; a width of the third FM layer is greater than a height of the third FM layer; and a width of the fourth FM layer is greater than a height of the fourth FM layer.

21. A magnetic sensor comprising the half bridge circuit of claim 15.
